

## SUMMARY

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- Research-oriented electronics engineer and Ph.D. candidate with +9 years of technical experience in astronomical instrumentation. Excellent technical writing, project management and mentoring skills. Extensive experience working in large collaboration with diverse teams of scientists and engineers.

## EDUCATION

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### Cornell University

Ithaca, NY

Ph.D. in Astronomy and Space Sciences.

2021–Present

- Thesis: “The Epoch of Reionization Spectrometer: Development and Early Science.” (in preparation).  
Expected completion: Spring 2027.

### Pontifical Catholic University of Peru

Lima, Peru

B.Sc. in Electronics Engineering. ABET accredited.

2011–2016

- Thesis: “Design of a LabVIEW-based polyphase filter bank spectrometer for radio astronomy using FlexRIO FPGA technology and CUDA-enabled GPU”.

## EXPERIENCE

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### CCAT Observatory

Ithaca, NY.

*Graduate Research Assistant.* Advisors: Prof. Gordon Stacey and Prof. Michael Niemack

2021 –Present

- Developed state-of-the-art submillimeter instrumentation for the CCAT collaboration.
- Leading the design, fabrication and deployment of a direct-detection imaging spectrometer.
- In charge of the development of thermometry and detector readout harnesses for the Prime-Cam receiver.

### Atacama Cosmology Telescope

San Pedro de Atacama, Chile.

*Site Engineer.* Advisors: Prof. Suzanne Staggs and Prof. Mark Devlin.

Oct 2019 –Jul 2021

- Ensured the continuous operation of complex astronomical instrumentation at very high altitude (17,000 ft). In charge of regular maintenance, troubleshooting and overhaul of electrical, cryogenic and mechanical systems.
- Performed calibration procedures like the alignment of the receiver with the primary and secondary reflectors through high-precision photogrammetry ( $\sim 10\mu\text{m}$ ). Carried out seasonal passband characterization via Fourier Transform Spectroscopy.
- Helped in the deployment of the last detector array upgrade (27/39 GHz) for the ACTPol receiver.

### Management Solutions

Lima, Peru. Madrid, Spain.

*Assistant Business Consultant*

Sep 2018 –Jul 2019

- Worked in big data and bank stress test for a major local bank in Lima, Peru. Software development and data science for a leading financial institution in Madrid, Spain.

### Institute for Radioastronomy

Lima, Peru.

*Research Engineer.* Advisor: Prof. Jorge Heraud

May 2017 –Aug 2018

- Worked in the construction of an 8-meter diameter radio telescope (RT8). Designed several hardware and software systems for the telescope’s electromechanical pointing system.
- Developed custom firmware for the backend electronics of the telescope.
- Assisted in the characterization of an 1.42 GHz feed antenna using software-defined radio technology.

## SKILLS

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- **Programming:** Python, C/C++, CUDA, LabVIEW, microcontroller firmware.
- **CAD/Simulation** Solidworks, KiCAD, Ansys Zemax Optics Studio.
- **Tools:** LaTeX, Git, Docker, Linux.

## AWARDS

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- Beca de Estímulo Académico Solidario (BEAS) 2013–2016  
*Fully-funded undergraduate scholarship for outstanding academic performance and financial need.*
- URSI Young Scientist Award Aug 2017  
*Merit-based registration fee waiver and travel grant to attend URSI 2017, Montreal, QC.*
- Summer School of Astronomical Instrumentation, Dunlap Institute, UofT Jul 2019  
*Merit-based tuition waiver and travel grant. Toronto, ON.*
- CASPER Workshop, Harvard & Smithsonian Center for Astrophysics (CfA) Aug 2019  
*Merit-based fee waiver and travel grant (invited, could not attend). Cambridge, MA.*

## TEACHING

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- **Teaching Assistant** at Cornell University Fall 2022, 2024, 2025  
*Multiwavelength astronomical techniques (ASTRO 4410)*
- **Teaching Assistant** at Cornell University Spring 2023  
*Our Solar System (ASTRO 1102)*
- **Teaching Assistant** at PUCP Aug 2017 - Dec 2017  
*Space Science and Engineering (ING306)*

## PUBLICATIONS

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- [1] B. Keller, **R. Freundt**, J. R. Burgoyne, S. Chapman, S. Choi, C. J. Duell, C. Groppi, C. Humphreys, L. T. Lin, A. Middleton, M. D. Niemack, D. Patel, E. Vavagiakis, S. Walker, Y. Wang, and R. Xie, “Ccat: Flexible stripline circuits for large-format kinetic inductance detector (kid) array readout”, *IEEE Transactions on Applied Superconductivity*, vol. 36, no. 6, pp. 1–6, 2026.
- [2] Middleton, A. et al. (including **R. G. Freundt**), “Ccat: Led mapping and characterization of the 280 ghz tin kid array”, *IEEE Transactions on Applied Superconductivity*, vol. 35, no. 5, pp. 1–4, 2025.
- [3] Vaskuri, A. K. et al. (including **R. G. Freundt**), “280-GHz aluminum MKID arrays for the Fred Young Submillimeter Telescope”, *Journal of Astronomical Telescopes, Instruments, and Systems*, vol. 11, 026005, p. 026 005, Apr. 2025.
- [4] **R. Freundt**, Y. Li, D. Henke, J. Austermann, J. R. Burgoyne, S. Chapman, S. K. Choi, C. J. Duell, Z. Huber, M. Niemack, T. Nikola, L. Lin, D. A. Riechers, G. Stacey, A. K. Vaskuri, E. M. Vavagiakis, J. Wheeler, and B. Zou, “CCAT: A status update on the EoR-Spec instrument module for Prime-Cam”, in *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII*, vol. 13102, SPIE, Aug. 16, 2024, pp. 343–352.
- [5] Collaboration, C.-P. et al. (including **R. G. Freundt**), “CCAT-prime Collaboration: Science Goals and Forecasts with Prime-Cam on the Fred Young Submillimeter Telescope”, *The Astrophysical Journal Supplement Series*, vol. 264, no. 1, p. 7, Dec. 2022, et al. including R. G. Freundt, ISSN: 0067-0049.

- [6] **R. Freundt** and J. A. Heraud, “Design of a LabVIEW-based polyphase filter bank spectrometer for radio astronomy using FlexRIO FPGA technology and CUDA-enabled GPU”, in *2017 XXXIInd General Assembly and Scientific Symposium of the International Union of Radio Science (URSI GASS)*, Aug. 2017, pp. 1–4.